10 Security Settings
**You Must Configure in**
# Windows 11

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1




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## Windows 11: Secure by default

*Our focused security work, driven by observing attacker patterns and behaviors, has resulted in a* ***reported 62% drop in security incidents***

*… in* ***comparison*** *with Windows 10 devices.*

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Windows security and resiliency: Protecting your business

4

Nicklas Ahlberg



rockenroll.tech

onevinn

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#1 Virtualization
# Based Security (VBS)

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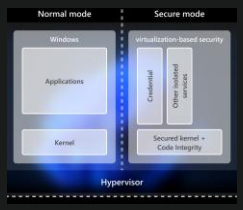
## Hardware requirements

- 64-bit CPU
- Second Level Address Translation (SLAT)
- IOMMUs or SMMUs (Intel VT-D, AMD-Vi, Arm64 SMMUs)
- Trusted Platform Module (TPM) 2.0
- Firmware support for SMM protection
- Unified Extensible Firmware Interface (UEFI) Memory Reporting
- Secure Memory Overwrite Request (MOR) revision 2
- Memory integrity-compatible drivers
- Secure Boot

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## # Virtualization-based security (VBS)




- Crucial for a good Windows security posture!
- Creates an isolated virtual environment for increased protection from vulnerabilities in the OS = the root of trust if the kernel has been compromised
- Protects security assets such as authenticated user credentials.

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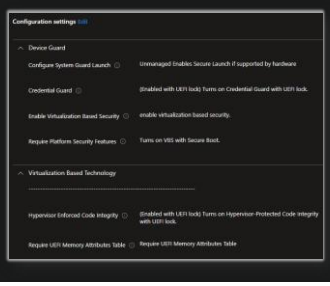
## # Virtualization-based security (VBS)




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## Enable VBS




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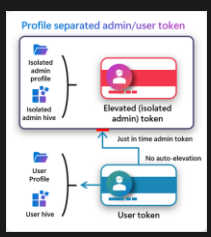
# #2 Administrator **Protection**

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## Administrator Protection

- **Just-in-time elevation:** Users remain deprivileged between admin tasks. Each task gets a new admin token, discarded after use.
- **Profile separation:** Hidden, system-generated accounts create isolated admin tokens, helping protect elevated sessions from user-level malware.
- **No auto-elevations:** Users must interactively authorize every admin operation.

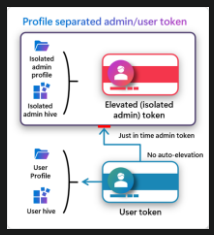



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## Administrator Protection

System Managed Admin Account (SMAA)
**User:** NicklasPC\Nicklas
**SMAA:** NicklasPC\ADMIN_Nicklas




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## Monitoring and Reporting

Administrator protection tracks elevations with two new Event Tracing for Windows (ETW) events.

**Existing provider: Microsoft-Windows-LUA** GUID {93c05d69-51a3-485e-877f-1806a8731346}

| Event ID | Event Name | Description |
|---|---|---|
| **15031** | Elevation Approved | Logged when a user successfully authenticates and elevation is granted |
| **15032** | Elevation Denied/Fail | Logged when elevation is denied, fails, or times out |

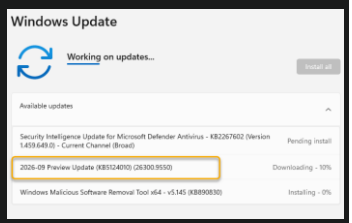
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## KB5124010 fixes key issues

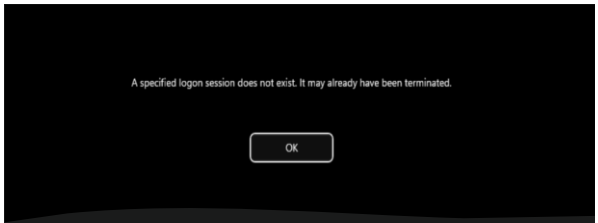
**Missing Start menu icons**
Fixes missing launch icons after some app installations.

**Unexpected elevation prompts**
Fixes unnecessary elevation requests from some apps using Edge extensions.




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Do not remove the **ADMIN_** account from local admins group.

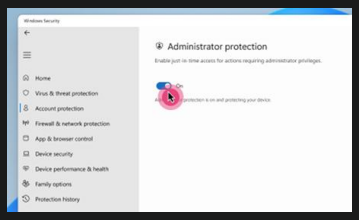
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## Something to look forward to…

**Windows Insider preview**
This option is currently available only in Windows Insider builds.

**Gradual rollout**
It will roll out to everyone gradually.




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## #3 Enhanced
## Sign-in Security (ESS)

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## Windows Hello

Researchers Reveal “Windows Hell No” Vulnerability in Windows Hello Biometric System

**Enhanced Sign in Security to the rescue!**

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## WHfB: Enhanced Sign in Security (ESS)

- ESS isolates the biometric data to trusted hardware or memory regions.
- Tamper protection!
- “Impossible” for malware to exploit the bio data to simulate a sign-in
- Utilizes VBS, so don’t skip the W11 HW requirements 🤩




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## Windows Hello for Business -Enhanced Sign-in Security

**Prerequisites**

- ❗ All the Windows 11 hardware requirements
- 📷Internal camera with ESS support
- 👆Internal fingerprint sensor with ESS support or **supported external HW!**
- 📄 BIOS with ESS support
- 💻It is clear that our OEM’s own the hardware requirements

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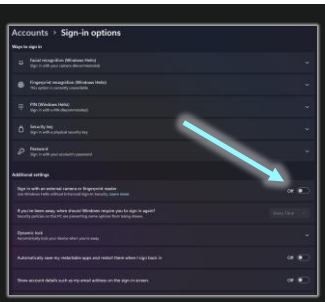
## External Fingerprint reader now supported!



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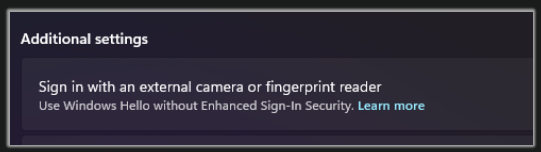
22

Settings ->
Accounts ->
Sign-in options



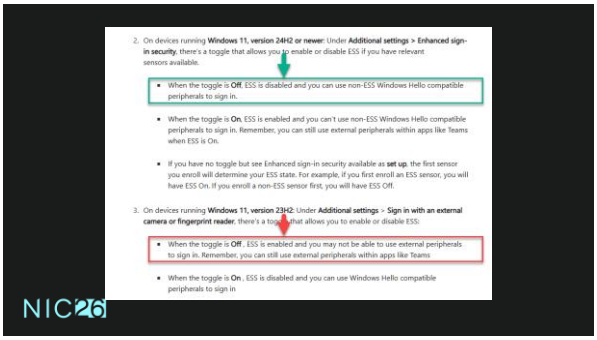

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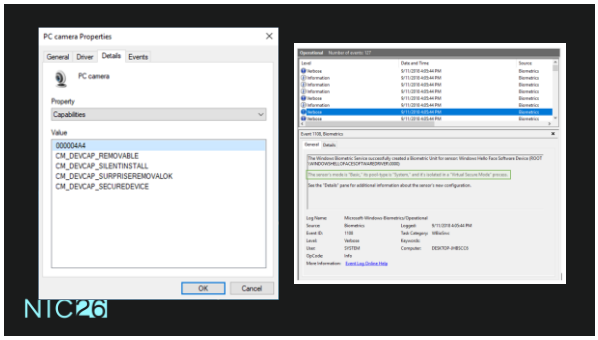



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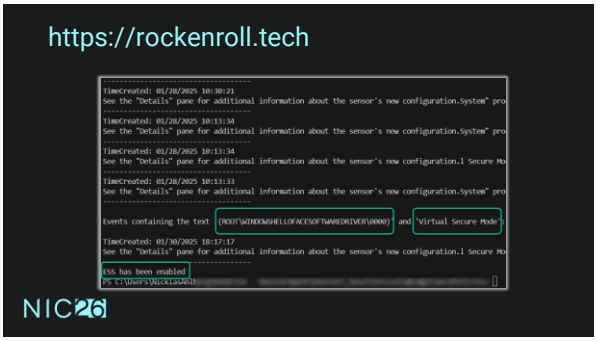
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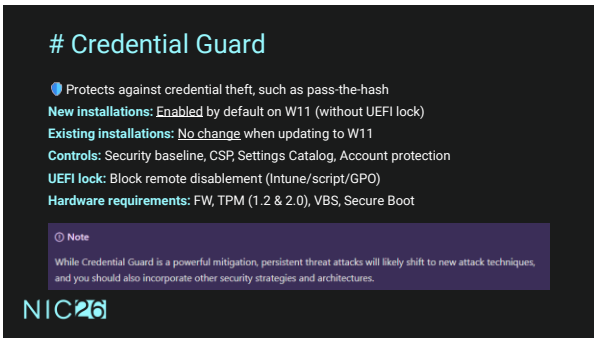

26



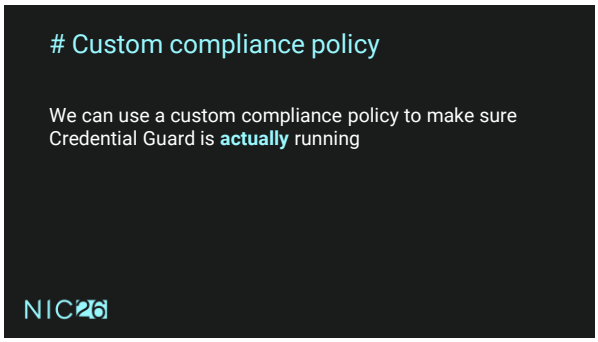

27



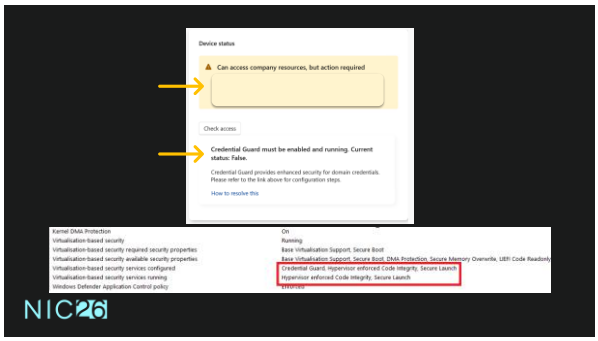

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#5 Endpoint
## Privileged Management

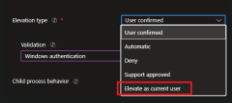
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## Endpoint Privilege Management

M365 E5 - Q3 2026

- Elevate current user is here!
- The number 1 challenge we had with developers, now they can elevate Visual Studio and still access code-repo, subscriptions and shares.



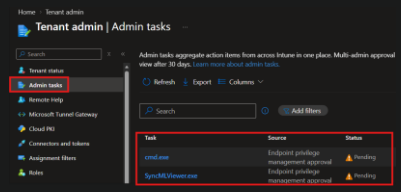

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## Endpoint Privilege Management

M365 E5 - Q3 2026

- Request elevation is now really fast
- From the user requests elevation it is visible in Intune Portal within seconds!
- Intune SR 2511 introduced Admin Tasks view which has potential




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#6 Microsoft
## Edge Management Our Most Important App?!

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## Microsoft Edge Security baseline

- Microsoft Edge Security baseline updated for version 151
- Microsoft Edge version 152 = release schedule every second week.
- This mean Security baselines for Microsoft Edge move to target Extended Stable in the future.

- *"To better align with the needs of the enterprise customers, the Microsoft Edge security baseline is expected to align with the, which provides major feature updates on an eight-week cadence while continuing to receive security and quality updates between releases"*

NIC26 https://techcommunity.microsoft.com/blog/microsoft-security-baselines/security-review-for-microsoft-edge-version-152/4550956

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## Microsoft Edge release cadence

- Starting with Microsoft Edge v.152 Stable channel is released with a 2 Week cadence.
- Extended Stable = 8 Weeks
- Which channel are you on?

| | | | | |
|---|---|---|---|---|
| 156 | Target release | 06-October-2026<br>150.0.3674.8 | Week of 22-October-2026 | Week of 22-October-2026 |
| 155 | Release version | 29-September-2026<br>150.0.3650.75 | 8-October-2026<br>153.0.4283.45 | Not applicable |
| 154 | Release version | 08-September-2026<br>150.0.3629.9 | 24-September-2026<br>154.0.4258.37 | Not applicable |
| 153 | Release version | 25-August-2026<br>150.0.3617.65 | 10-September-2026<br>153.0.4216.32 | Not applicable |
| 152 | Release version | 06-August-2026<br>150.0.3591.55 | 27-August-2026<br>152.0.4191.53 | 27-August-2026<br>152.0.4191.53 |
| 151 | Release version | 10-July-2026<br>150.0.3579.75 | 21-July-2026<br>151.0.4135.58 | Not applicable |
| 150 | Release version | 11-June-2026<br>150.0.3579.5 | 02-July-2026<br>150.0.4078.89 | 02-July-2026<br>150.0.4078.80 |

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## Microsoft Edge Security baseline v.151

- 6 recommended settings:
  - Enable Process Isolation
  - Enable renderer in app container
  - Enable the network service sandbox
  - Configure browser process code integrity guard
  - Enable Application Bound Encryption
  - Enhance the security state in Microsoft Edge: Balanced
- **Important:** Enable Process Isolation

"If this policy is unset, Microsoft Edge will follow the default rollout process for the Process Isolation feature, which means that the feature will be gradually rolled out to an increasing number of users."

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## Edge Extensions: Security

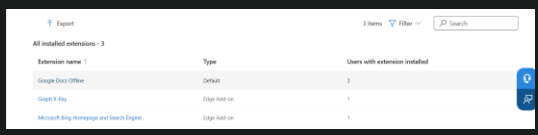
Malicious GhostPoster browser extensions found with 840,000 installs




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## Edge Admin Service: Extension Monitoring




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## #7 Windows 11
# Memory Integrity

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## Memory Integrity: Automatically Enabled!! 2026-10

- Built on Virtualization-based Security (VBS).
- Helps protect critical parts of Windows from tampering.
- Hotpatch updates.
- **Disable** Memory Integrity will be honored.
- HVCI protects against common attacks like WannaCry that rely on the ability to inject malicious code into the kernel.



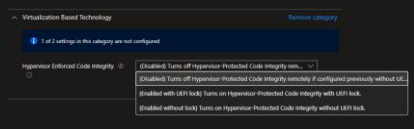

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Expanding memory integrity protection across Windows devices - Windows IT Pro Blog

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## Memory Integrity: Automatically Enabled!! 2026-10

- Gaming/R&D/CAD: about 15% CPU loss
- Anti-cheat software may block you from gaming!!!! 🎮




Expanding memory integrity protection across Windows devices - Windows IT Pro Blog

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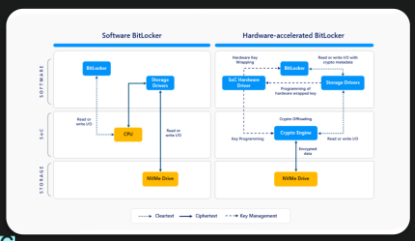
43

## #8 Hardware-accelerated **BitLocker**

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## Hardware-accelerated BitLocker

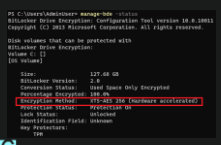


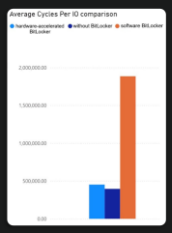

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## BitLocker: Crypto offloading

- Move BitLocker cryptographic operations from the CPU to a dedicated crypto engine.
- This helps improve both performance and battery life.




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## BitLocker: Hardware protected keys

- BitLocker bulk encryption keys are hardware wrapped.
- This increases security by reducing their exposure to CPU and memory vulnerabilities.

```
BitLocker on                    : TRUE
Algorithm                       : XTS-AES256
HwCryptoOffloadingSupported     : TRUE
HwCryptoOffloadingUsed          : TRUE
HwWrappingSupported             : TRUE
HwWrappingUsed                  : TRUE
HwOffloadType                   : NVMe
```

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## Hardware-accelerated BitLocker

- NVMe drive
- XTS-AES-256 algorithm




New Intel Core Ultra Series 3 Processors




"Systems will be available globally starting Jan. 27, 2026"

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#9 Device Control
## Block USB Storage

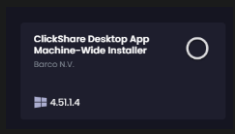
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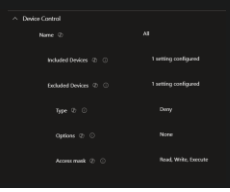
49

## Device Control

- USB devices are used less than ever with all cloud storage solutions, but we must still control it!
- We can still allow what is required by the business side.

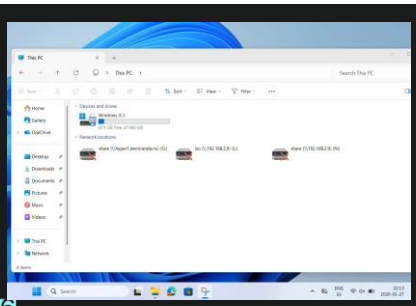
Tip: Some devices like Barco Clickshare can become a challenge if not allowed.




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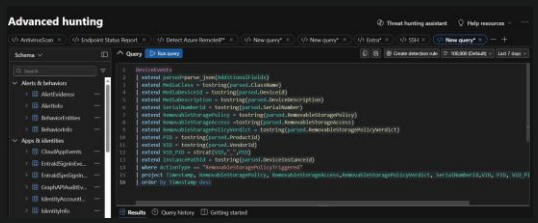
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## Advanced Hunting




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#10 Multifactor
## Unlock

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## Windows Hello for Business: Multi-factor Unlock

- Require another factor to sign in
- Shoulder surfing
- Passwords and security key will not require MFU
- Supports: PIN, Facial, Fingerprint and passive (connected to a specific network or Bluetooth device)



NIC26 Rockenroll.tech Windows Hello for Business - MFU

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## Summary

- # VBS
- # Credential Guard and Custom Compliance
- # EPM - Latest features
- # Microsoft Edge
- # Enhanced Sign-in Security
- # Admin Protection
- # Hardware-accelerated BitLocker
- # Device Control: Block USB
- # Memory Integrity
- # WHfB: Multifactor Unlock
- # BitLocker PIN (Bonus)

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## #11 BitLocker **PIN!!**

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- Protects against DMA port attacks before Windows has booted
- We see more and more customers using BitLocker PIN
- Setting a PIN after Windows provisioning is challenging
- PIN is part of the bitpixie mitigations strategy

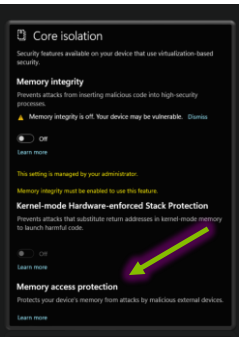



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## **(Bonus)** Kernel DMA Protection

- Protects against DMA attacks, **after** Windows has booted
- Drive-by DMA attacks
- Requires the device/driver to support DMA-remapping (isolated space within memory)
- Enabled by default, if **UEFI** has been configured correctly




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